

NEXA GENOMICS INSTITUTE

SYNAPSE BIOINFORMATICS CORE PLATFORM

Computational Bio-Pipeline Run Log & Execution Audit Trail

RUN COMPLETED

Generated: 2026-07-24 04:42:15 UTC

Pipeline: WGS-GATK4-v3.4.1

Execution Metadata

Run ID: RUN-2026-0724-WGS-9842
Project: PRJ-NEXA-2026-HG002
Sample ID: HG002-NIST-NA24385
Lead Analyst: Dr. Elena Rostova, PhD
Compute Node: node-042.cluster.synapse.bio
Scheduler: SLURM Job ID #8492014
Container: docker://synapsebio/wgs:3.4.1

Run Performance Metrics

Start Time: 2026-07-24 00:29:27 UTC
End Time: 2026-07-24 04:42:15 UTC
Total Wall Time: 04h 12m 48s (15,168 s)
CPU Allocation: 32 Cores (Xeon 8380)
Memory Peak: 46.2 GB / 128 GB Requested
Storage I/O: 240.5 GB Read / 185.2 GB Write
Exit Status: 0 (Clean Exit)

1. Workflow Architecture & Pipeline DAG

The WGS-GATK4 germline variant calling pipeline ingests high-coverage paired-end Illumina NovaSeq FASTQ files and produces recalibrated BAM files alongside high-confidence single nucleotide variants (SNVs) and small Indels in standard VCF/gVCF format following GATK Best Practices.



2. Software Manifest & Dependency Specifications

All pipeline steps were executed within an immutable Docker container. Table 1 details binary tool versions, execution paths, and parameters utilized.

Tool Name	Version	Binary Location	Key Execution Flags
fastp	v0.23.4	/opt/conda/bin/fastp	-detect_adapter_for_pe -thread 16 -q 20
bwa-mem2	v2.2.1	/opt/conda/bin/bwa-mem2	mem -t 32 -R 'RG\tID:RUN9842\tSM:HG002'
samtools	v1.19	/opt/conda/bin/samtools	sort -@ 8 -m 4G -o output.bam
GATK4	v4.5.0.0	/opt/gatk/bin/gatk	HaplotypeCaller -native-pair-hmm-threads 16
bcftools	v1.19	/opt/conda/bin/bcftools	stats -F GRCh38.fa -s HG002
multiqc	v1.21	/opt/conda/bin/multiqc	-force -export -title "WGS Run 9842"

Table 1: Complete software inventory for pipeline execution run RUN-2026-0724-WGS-9842.